Capturing Temporal Dynamics in Time Series Data with Generative Adversarial Networks (GANs)

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This paper is saying in a very full way that we are looking at Generative Adversarial Networks (GANS) for how they make the moving time things in time series data and we are seeing how this adversarial learning thing is catching the very mixed and long chain type dependence in many places like healthcare finance iot and cyber security and we are telling one big simple way to understand all the temporal gan types and we are checking the benchmark methods and we are also showing the problems that come like stability and understanding the model and keeping the data safe and private and we are giving one clean summary of all the big works from 2014 to 2025 and we are saying what can come next in the new generative models that are like foundation models and bro after that to make the gap small between theory and real world we are also doing one case study with Timegan on the s and p 500 data and with this we are showing that the evaluation numbers we talked about are working and we are showing that this model can catch the very twisty and moving time patterns even when the market is going up and down too much

ACM Reference Format:
Muhammad Mustafa Khan, Muhammad Saqib Kakar, and Muhammad Umar Khan Lodhi. 2025. Capturing Temporal Dynamics in Time Series Data with Generative Adversarial Networks (GANs). *ACM Comput. Surv.* 1, 1, Article 1 (November 2025), 23 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

1.1 Motivation

Time series data is holding very important decision making in places like healthcare finance IoT and cyber security and in real life we see that temporal data is many times broken and noisy and also we cannot share it freely because of privacy and this makes learning the time changing pattern very important and also very hard and when we make fake time series data with gans then we get help because we can make more data for training and we can keep privacy and we can also make simulation that looks real for many moving variables [2, 35].

The exponential growth of temporal data in many industries is showing that we really need strong generative frameworks and people are saying that more than 60 percent of enterprise data will be time indexed by 2025 and this is in many areas like clinical sensors and financial ledgers and cyber security logs [11, 76] and still only a small part of this data is public because of privacy rules like gdpr and hipaa and this makes big problems for sharing and for repeating the work and for making models that can work well everywhere [53, 79]. Consequently, temporal generative modeling—particularly adversarial approaches—has become central to data augmentation, anomaly

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ACM 1557-7341/2025/11-ART1
https://doi.org/10.1145/nnnnnnn.nnnnnnn

simulation, and digital-twin construction [62, 138]. While earlier surveys have examined general generative frameworks [15, 67], none provide a systematic synthesis of adversarial, hybrid, and diffusion-augmented temporal GANs across domains and tasks. This survey addresses that gap by unifying theoretical, architectural, and application-level perspectives within a comprehensive cross-domain taxonomy.

1.2 Problem Statement and Research Gap

Traditional models such as ARIMA, RNNs, and LSTMs capture trends and short-term dependencies but fail to replicate multimodal, stochastic, or non-stationary temporal patterns [4, 22, 29]. GANs overcome this by learning implicit probability distributions via adversarial training between a generator and discriminator [2, 35, 106]. However we still face many problems because it is very hard to catch the proper temporal flow and keep the model stable and also it is difficult to check the quality in different domains

1.3 Survey Objectives and Contributions

- **Unified Taxonomy:** we are giving one simple classification of temporal gan models by their architecture and how they are used in different domains.
- **Comparative Analysis:** we are checking the training goals and the privacy methods in places like healthcare finance and iot.
- **Empirical Validation:** we are doing one case study where we use timegan on financial data and we show how the common evaluation methods like t sne and tstr are working.
- **Future Roadmap:** we are showing the problems that still stay like instability and causality and we are giving some research directions for the coming years.

1.4 Paper Organization

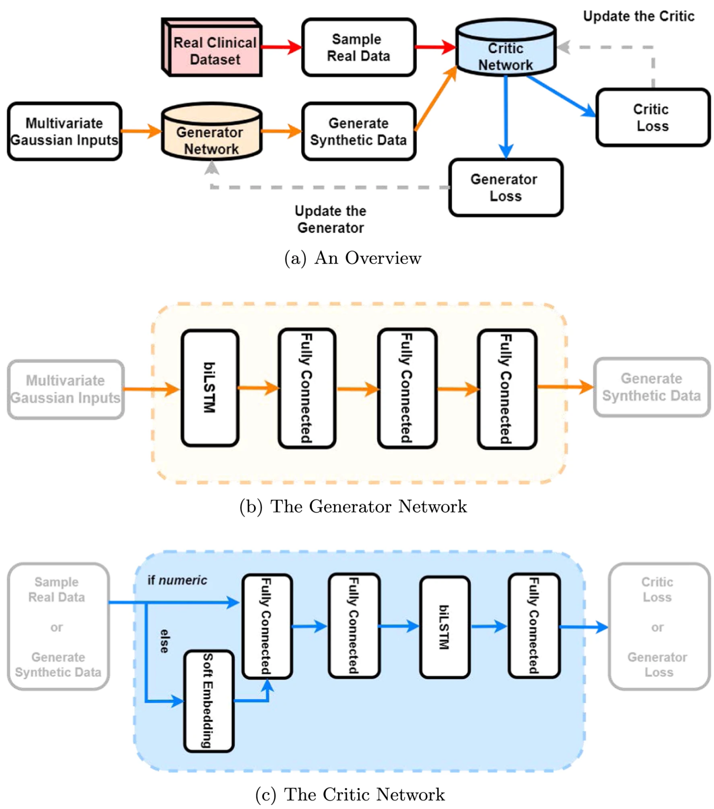
Section 2 is talking about the basic theory that we need and section 3 is showing how gans for time series changed step by step over the years and section 4 is giving the taxonomy and the comparative evaluation in a simple way and section 5 is showing how these models are used in many domains and section 6 is giving a deep look at how people check and judge the temporal generative models and section 7 is the empirical case study where we use timegan on the s and p 500 financial data and section 8 is pointing out the open problems and section 9 is closing the paper with the final message.

2 Background and Theoretical Foundations

2.1 Generative Adversarial Networks

Generative Adversarial Networks (GANS) introduced by goodfellow et al. in 2014 [35] are bringing a big change in how we do generative modeling and unlike the old probabilistic ways like variational autoencoders vaes or maximum likelihood estimation [34, 57] We see that gans try to learn the very complex and high dimensional data distributions by making two neural networks fight each other and these two networks are a generator G and a discriminator D and the generator is trying to make fake samples that look like the real data and the discriminator is trying to catch which sample is real and which one is fake

This minimax game is pushing G and D to learn together and this helps them catch many types of multimodal and non linear patterns in the data [2, 89, 106] and later many people made this system stronger by adding better loss functions like wasserstein and least squares [38, 86] and also by adding methods that make the model more stable like spectral normalization and gradient



(a) An Overview

(b) The Generator Network

(c) The Critic Network

Fig. 1. Conceptual overview of a temporal GAN pipeline showing data flow from real sequences to latent representations and generated time series. Reproduced and adapted from Kuo et al. (2022) [62].

penalties [88, 92] and because of all this work gans have become a very flexible generative tool in deep learning used in many places from images and audio to time series [15, 76]

Gans have done very strong work in static areas like image creation natural language modeling and making more training data for structured datasets [2, 89, 106] and the big power of gans is that they can learn multimodal and non linear and non gaussian distributions without needing hard assumptions about the data likelihood

2.2 Mathematical Formulation

Formally the generator $G(z;\theta_g)$ is taking a noise vector $z \sim p_z(z)$ and mapping it to a fake sample $x_{fake} = G(z)$ and the discriminator $D(x;\theta_d)$ is giving a probability for whether some input x is real or fake and the adversarial training goal is written as

$$\min_G \max_D V(D,G) = \mathbb{E}x \sim pdata(x)[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \tag{1}$$

and when we optimize this D is trying to become better at catching real and fake data and G is trying to fool D and because of this feedback loop the generator starts making more and more real looking samples until it gets close to the true data distribution $p_{data}(x)$ [2]

2.2.1 Temporal Consistency and Sequence Dependencies. Unlike image data where the spatial relations are mostly local we see that temporal sequences have long range and causal connections and this breaks the simple independent assumption that normal gans use and to make the time flow consistent many works add recurrent layers inside the generator or the discriminator and one early example is mogrens c rnn gan [93] which used recurrent networks to catch the sequence smoothness and yoon et al. [138] added supervised embedding losses so that the latent movement matches the real transitions and later methods like cot gan [151] and temporal causal gan [141] make the sequence generation more formal by using causal optimal transport and interpretable latent paths and all these changes together help fix problems like mode collapse and phase mismatch that often happen in temporal data generation.

2.3 Extensions to Temporal Modeling

While classical GANs were designed for i.i.d. (independent and identically distributed) samples such as images, many real-world problems require the generation of **sequential or time-dependent data**. Extending GANs to such domains introduces the challenge of capturing temporal dependencies and continuity between successive samples [29, 138].

To address this, recurrent architectures and embedding networks were incorporated into the generator and discriminator. For instance, the **C-RNN-GAN** model replaced fully connected layers with LSTMs to generate sequences of music and signals [93]. Subsequently, the **Recurrent Conditional GAN (RCGAN)** [29] introduced conditional inputs (e.g., patient records) to ensure contextual consistency. The **TimeGAN** architecture [138] unified adversarial learning with supervised temporal embedding, combining autoencoding and generative objectives to preserve temporal and contextual correlations.

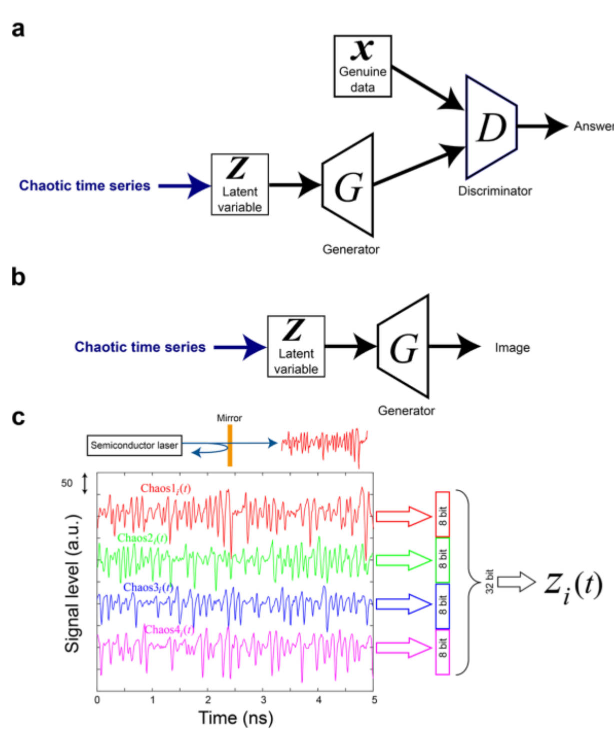


Fig. 2. Basic architecture of a Generative Adversarial Network (GAN) adapted for time-series reconstruction. The generator (G) maps latent variables Z to synthetic sequences, while the discriminator (D) distinguishes generated data from real observations. This is reproduced and adapted from Jiang et al. (2019) [50].

3 Evolution of Temporal GANs (2014–2025)

The development of generative adversarial networks gans for time series data has moved slowly from simple adversarial learning on static data to more advanced models that can catch very complex temporal patterns and in this section we are giving one timeline style overview from the early recurrent adversarial ideas to the hybrid conditional and new foundation model based frameworks

3.1 From Static GANs to Temporal Modeling (2014–2017)

The first gan model by goodfellow et al. [35] made a two player game between a generator and a discriminator and even though these early gans were very good for image making they could not understand time because they were only for static data and works like dcgan [106] and wgan [2] made training more stable but they still stayed in the i i d world

To bring time information mogren introduced the *continuous recurrent gan c rnn gan* [93] Where the normal feed forward layers were replaced by rnn cells and this helped both the generator and discriminator to see the time flow and soon after esteban et al. [29] made the *recurrent conditional gan rcgan* for making realistic medical time series which showed that adversarial training can work nicely for temporal data making

3.2 Hybrid and Conditional Temporal GANs (2018–2020)

Between 2018 and 2020 the research started to move toward hybrid models that learn both the time smoothness and the context meaning together and the *time series generative adversarial network timegan* by yoon et al. [138] became a very big step in this journey and unlike the earlier models timegan mixed supervised unsupervised and reconstruction losses inside one adversarial setting and this helped it keep both the time dependency and the hidden meaning of the data

timegan brought four parts working together an *embedding network* that encodes the real sequences a *recovery network* that rebuilds them a *generator* that makes new latent sequences and a *discriminator* that checks real and fake and this mixing of supervised and adversarial training made the generated sequences much more real looking especially in healthcare finance and iot

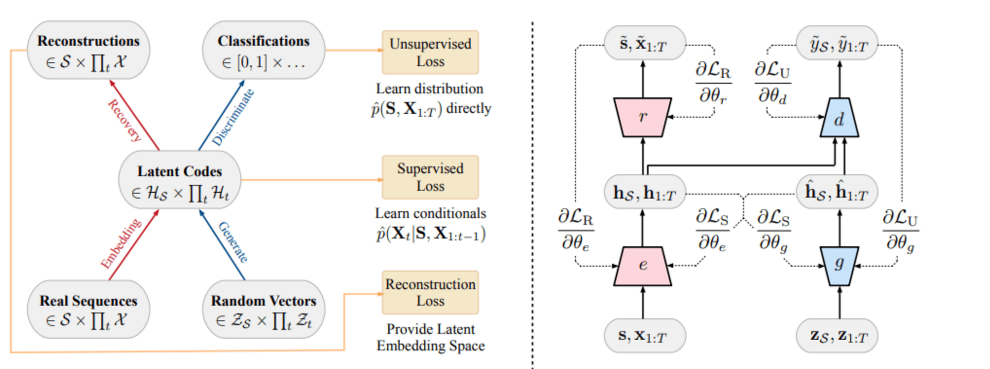


Fig. 3. conceptual block diagram and training scheme of the timegan architecture showing how we have the embedding part the recovery part the generator and the discriminator and how they all train together with supervised and unsupervised losses and this is adapted from yoon et al. (2019). [138].

After this more models came like conditional optimal transport gan cot gan [151] and wasserstein timegan versions [2] which gave more stability by using distance based ideas and transformer style training and these improvements made sequence control and detailed temporal conditioning easier

3.3 Domain-Specific Temporal GANs (2020–2023)

By 2020 many domain specific temporal generation models started to appear in healthcare kuo et al. [62] introduced *healthgym* which is a synthetic patient data system inspired by timegan but made stronger for privacy and reinforcement learning and in finance models like *fingan* and sensor focused systems like *tts gan* [49] showed that domain tuned losses help a lot when the data is non stationary or highly multivariate

These models also showed a shift from pure rnn based models to transformer gan hybrids where attention helps to learn long range time links and table 1 is giving a summary of the important architectures and what they contributed

Table 1. Representative Temporal GAN Architectures (2016–2023)

Model	**Year**	**Architecture**	**Domain**	**Contribution**
C-RNN-GAN	2016	RNN + GAN	Music	Sequential data generation
RCGAN	2017	Conditional RNN + GAN	Healthcare	Correlation preservation
TimeGAN	2019	Hybrid (Embedding + GAN)	Multi-domain	Temporal consistency
COT-GAN	2020	Optimal Transport GAN	Forecasting	Improved stability
TTS-GAN	2021	Transformer + GAN	IoT	Long-term dependency modeling
HealthGym	2022	Differentially Private GAN	Healthcare	Privacy-preserving synthesis

3.4 Diffusion-Era and Foundation-Model Temporal GANs (2023–2025)

Recent years are showing that adversarial learning and diffusion based generative modeling are coming close together and methods like *diffusion ts* [97] and *aigc ts* [149] are mixing temporal attention with probabilistic diffusion so we can get better long horizon forecasting and hybrid models like *metaindux ts* [136] are joining foundation model pretraining with gan based fine tuning and this is making one new way for large scale and cross domain time series generation

These improvements are telling us that future temporal gans will move toward multi modal foundation level systems that can keep learning all the time and can also do zero shot generalization across many domains

Overall this ten year journey is showing a clear shift from the early hand made recurrent adversarial models [93] to the new self supervised transformer based [49, 122] and diffusion combined designs [45, 97, 136] and this is putting temporal gans as an early step toward foundation level generative models [151] and this big change is showing that deep generative modeling is moving from architecture focused thinking to representation focused thinking where adversarial learning optimal transport and probabilistic diffusion all work together inside shared latent spaces

As transformer and diffusion methods get more mature we see that temporal gans keep helping in the design of scalable understandable and domain free time series generators that mix statistical strength with foundation model flexibility

Building on the evolution story explained in section 3 we now bring together the wide work on temporal gans into one clean taxonomy that shows the method level architecture level and objective level dimensions.

4 Taxonomy and Classification of Temporal GAN Models

4.1 Overview

While early generative adversarial networks gans were mostly made for static things like image generation [35, 106] later work made a big family of **temporal gan architectures** that can model dynamic moving and multivariate time series data and because we have many architectures and many goals and many domains it becomes important to organize this work in one clean taxonomy and following the way previous surveys on time series generation did [15, 67, 76] this section is classifying temporal gans using three axes which are 1 **data regime** 2 **architectural backbone** and 3 **training objective**

Unlike older surveys that only show one big taxonomy figure we are giving an analytical comparison that maps the important models in a clear way and this is shown in table 2 which gives transparency and helps readers find the correct references for each model and also keeps the scientific standard strong

4.2 Classification Dimensions

(1) Data Regime: Discrete vs. Continuous Time-Series. the first axis is about whether the data is discrete event sequences like symbolic music transaction logs or healthcare event codes or if it is continuous values like physiological signals stock movements or sensor readings and this idea comes from brophy et al. [15] because these two types have different assumptions sampling styles and loss designs and discrete models usually use categorical embeddings or seq2seq structures while continuous models use differentiable regression and noise based objectives

(2) Architectural Backbone. Architecturally temporal gans extend the normal generator and discriminator by adding time modeling parts like recurrent neural networks rnns [29**?**] or temporal convolutional networks tcns [4] or hybrid autoencoder gan layouts [138] or transformer based attention [135, 150] or score based diffusion models [45, 80] and these help in learning short memory long range dependence and non stationary patterns

(3) Training Objective. Temporal gans use many adversarial objectives not only the old jensen shannon divergence and some common ones are wasserstein loss with gradient penalty wgan gp [2, 38] or maximum mean discrepancy mmd [68] or causal optimal transport cot [135] and hybrid losses like in timegan [138] mix supervised reconstruction with adversarial training to make the temporal embeddings stable and reduce mode collapse and recently diffusion based models are becoming popular because they stay stable for industrial time series [80, 125]

4.3 Representative Model Families

table 2 is giving the main temporal gan families and shows their data type objective backbone and where they are used and this comparison shows how the field moved from simple rnn based adversarial designs to hybrid latent attention driven and diffusion powered models which often add domain knowledge and self supervised representations

Table 2. Comparative taxonomy of representative Temporal GAN architectures, categorized by data regime, adversarial objective, architectural backbone, and primary application domain.

Model Family	Data Regime	Adversarial Objective	Architectural Backbone	Primary Domains / References
C-RNN-GAN [93]	Discrete	JS Divergence	LSTM-based Sequence Generator	Symbolic music, event prediction
RCGAN [29]	Continuous	Conditional JS Divergence	Recurrent Conditional Generator (RNN)	Healthcare records, multivariate signals
TimeGAN [138]	Continuous	Hybrid (Supervised + Adversarial)	Autoencoder + LSTM + Embedding Network	Financial, IoT, human activity data
WGAN-GP [38]	Continuous	Wasserstein Distance	CNN / RNN Generator–Discriminator	Energy demand, electrocardiogram (ECG)
COT-GAN [135]	Continuous	Causal Optimal Transport	Transformer-based Sequence Modeling	Forecasting, motion trajectories
TTS-GAN [150]	Continuous	JS Divergence + Attention Consistency	Transformer Encoder–Decoder	Weather, traffic, climate modeling
Diff-MTS [80]	Continuous	Diffusion / Score-based Loss	U-Net Temporal Backbone	Industrial process monitoring
AIGC-TS [125]	Continuous	Generative Diffusion + Adversarial Refinement	Foundation-scale Transformer	Industrial IoT, anomaly simulation

4.4 Discussion

This taxonomy is showing some big trends and first we see that the field moved from only recurrent adversarial models to hybrid latent supervised models that mix autoencoding and adversarial loss and second attention and transformer systems are becoming common because they handle long range and cross variable links much better than old rnn based gans and third diffusion methods even though they are different from adversarial training are slowly merging with gan style workflows in industrial and foundation model settings [76, 125] and finally domain specific versions in healthcare [15, 29] finance [138, 147] and energy [80] show that good temporal generation needs a balance of realism diversity and easy interpretation under real world limits

From the method side this taxonomy also shows a merging of adversarial optimal transport and diffusion ideas inside time series generation and new works mix adversarial learning with probabilistic and self supervised losses to get more variety and stability [45, 97, 151] and models using transformers combine sequence level contrastive pretraining with adversarial fine tuning for stronger temporal reasoning and cross domain work [49, 122] and diffusion based systems like diff mts [80] and aigc ts [149] show how denoising and adversarial refinement can join to make long horizon generation more stable and meta pretrained systems like metaindux ts [136] show that we are moving toward foundation level generators that can adapt with very little data

Overall the taxonomy here is giving one unified view of how temporal gans have grown and it acts like a bridge from the model classification in this section to the benchmarking and application parts that come next

5 Applications of Temporal GANs

Generative adversarial networks gans have changed how we make check and use synthetic temporal data in many places where privacy scarcity or irregular sampling make model training difficult [15, 67, 76] and temporal gans let us make sequences that look real while still keeping the important statistical and causal properties of the real world [111, 135, 141] and now these models are used not only in research but also in serious areas like healthcare finance and cyber security where privacy safe data generation and strong simulation are very important [11, 53, 126] and this section is looking at domain specific applications in healthcare finance iot and smart city systems cyber security and human centered media and for each area we are giving the main architectures datasets and important findings

5.1 Healthcare and Biomedical Time-Series

The healthcare field has gained a lot from temporal gans because clinical data is both sensitive and sparse and early works like the *recurrent conditional gan rcgan* [29] showed that we can make realistic ecg and intensive care signals using adversarial training and later baowaly et al. 2019 made *medgan* and *ehrgan* [6] to generate synthetic electronic health records with focus on privacy safe data augmentation and kuo et al. 2022 brought *healthgym* [62] which mixed reinforcement learning environments with timegan style generation for critical care decision making

Recent work is also going into multi modal bio signals and models like *ecg gan* [24] and *bioseq-gan* [65] use convolution and recurrent hybrids to make more realistic morphology and privacy focused versions like *dp timegan* [79] and other local differential privacy models [123, 134] help follow hipaa and gdpr rules

benchmark projects like *timeseriesganbench* [56] and *tgebench* [78] are giving standard tests for physiological realism and cross patient generalization using datasets like physiOnet and mimic iii and recent studies also try to add explainable and causal parts into medical gans [111, 141] so that the latent movement can be connected to clinical meaning

Overall these works show that temporal gans can increase limited clinical data help with privacy safe sharing and support continuous learning in personalized medicine but the quality checks still differ across studies and many generated samples do not fully match real physiology

5.2 Financial and Economic Data

Financial time series are difficult because they are non stationary and have sudden regime changes and heavy tails and *timegan* [138] and related models have become basic tools for market simulation and trading and zhang et al. 2020 made *stockgan* [147] to generate high frequency stock movements and wiese et al. 2020 used wgan gp for volatility and derivative pricing [127]

Later systems like *fingan* [72] and *crypgan* [41] added conditional and attention based ideas to make multivariate financial sequences across many asset classes and zhang et al. 2023 proposed the *tfid* metric [140] for checking realism and *quantdiff gan* [117] mixed diffusion with adversarial learning for forecasts that take volatility into account and models like *findiff ts* [124] and *fedfingan* [134] support privacy safe financial data generation

Causal modeling is also being used to explain market changes inside the latent space [111, 141] and benchmark tools like *tgebench* [78] give common tests for generative models in financial forecasting

Many studies show that these models can copy autocorrelation and tail risk patterns giving synthetic but realistic data for back testing and risk work but still regulators worry about leakage and lack of explanation [53, 126]

5.3 IoT, Energy, and Smart-City Systems

Temporal gans play an important role in iot and cyber physical systems where sensor data can be noisy incomplete or private and jeon et al. 2021 made *tts gan* [49] for realistic traffic sensor sequences helping congestion prediction and zhou et al. 2022 improved temporal consistency with transformer based attention in *tts ganv2* [150]

in energy systems models like *e gan* [21] *powergan* [71] and *griddiff gan* [46] generate load and renewable profiles for grid analysis and *diff mts* [80] uses diffusion for industrial fault diagnosis and hybrid transformer gan ideas like *smartcitygan* [31] and *urbandiff ts* [148] join adversarial and score based training for large scale urban simulation

federated systems like *fedtimegan* [134] and privacy enhanced models [77, 123] allow sharing across organizations without giving raw data keeping security and data ownership safe

these synthetic sequences help in anomaly detection [108] energy optimization [71] and digital twin simulation for smart city planning [118] and show that gans can work well even for real infrastructure problems

5.4 Cybersecurity and Network Analytics

Adversarial models help a lot in cyber security because they can simulate rare but dangerous network events and lin et al. 2023 made *netgan ids* [77] for intrusion detection benchmarking and rathore et al. 2022 proposed *attackgan* [108] to imitate coordinated attacks to strengthen anomaly detectors under data imbalance

New hybrid transformer–gan systems [20] generate logs that keep long term dependence between system calls and network states and privacy safe models like *dp cybergan* [123] hide sensitive metadata and diffusion–adversarial models like *secdiff gan* [74] make rare intrusion patterns and *fedids gan* [146] supports federated training across organizations

Benchmark tools like *cic ganbench* [133] help with standard evaluation using public datasets like nsl kdd cicids2017 and ton iot and recent survey work [53, 69] warns about issues like explainability bias and responsible use of these synthetic data

Overall these methods not only grow the training data for ids but also show weaknesses in detection systems by adversarial stress testing and privacy aware simulation

5.5 Human Motion, Speech, and Multimodal Media

Temporal gans have also helped in creative and interactive areas like human motion speech and multimodal media and mogren first used *c rnn gan* [93] for symbolic music showing that recurrent adversarial training can make sequential creative outputs

later *dancegan* [66] and *posegan* [16] made realistic dance and gesture sequences using audio cues and this inspired more work on embodied generative modeling and transformer based models like *t2m gan* [101] *audio2gestures* [33] and *moglow gan* [44] improved expressiveness and control for text to motion and audio conditioned gestures

In speech and audio *wavegan* [26] and *specgan* [28] brought adversarial training to raw audio and diffusion–adversarial hybrids like *voicediff gan* [98] *diffwave gan* [60] and *uniaudiogan* [47] allow expressive multilingual and controllable tts

large multimodal systems like *aigc ts* [149] and *multimodal tsformer* [80] now combine audio text vision and motion for unified generation and benchmarks like aist++ [70] lrs3 [1] and humanml3d [39] help test temporal coherence and perceptual quality these models support creative industries human computer interaction virtual avatars and immersive analytics showing how wide and scalable temporal gans have become in the aigc ecosystem

5.6 Summary of Domain Trends

Across all domains, temporal GANs have shifted from proof-of-concept prototypes to production-level components for simulation, augmentation, and privacy-conscious data sharing. Three trends emerge: (1) domain-specific loss functions are essential to align adversarial training with application semantics; (2) hybrid diffusion-GAN architectures enhance temporal stability; and (3) privacy-aware and federated variants will be pivotal for real-world deployment. Table 3 summarizes representative studies and application categories.

Table 3. Representative applications of Temporal GANs across domains (2017–2025).

Domain	Representative Models	Key Contributions / Notes
Healthcare	RCGAN [29], medGAN [6], DP-TimeGAN [79], ECG-GAN [24], ehrGAN [6], HealthGym [62]	Privacy-preserving synthetic EHR and physiological signal generation
Finance	TimeGAN [138], StockGAN [147], Quant-GAN [127], FinGAN [72], CrypGAN [41], FinDiff-TS [124], GAN-Finance [129]	Realistic market simulation, volatility forecasting, and risk modeling
IoT / Energy	TTS-GAN [49], TTS-GANv2 [150], E-GAN [21], PowerGAN [71], Diff-MTS [80], FedTimeGAN [132], GridGAN [94]	Synthetic sensor and load data for smart-city and industrial systems
Cybersecurity	AttackGAN [108], NetGAN-IDS [77], DP-CyberGAN [123], LogGAN [137], Flow-GAN [19]	Adversarial simulation for intrusion detection and anomaly generation
Human Motion / Speech	C-RNN-GAN [93], DanceGAN [66], PoseGAN [16], AIST++ [70], SpecGAN [28], WaveGAN [26], VoiceDiff-GAN [98], HumanML3D [39], LRS3 [1]	Coherent human-motion and speech sequence synthe- sis across modalities

In summary, the growing diversity of applications underscores the adaptability of adversarial temporal modeling. Yet, consistent evaluation protocols, interpretability, and ethical governance remain open challenges, motivating the next section on benchmarking and evaluation.

6 Benchmarking and Evaluation of Temporal GANs

Evaluating Temporal GANs is much harder than evaluating normal image generation because time-based data must keep not only the individual values correct, but also the relationships between those values over time. Unlike benchmarks for images – which mostly depend on simple metrics like FID or IS [12, 83]—evaluating temporal data is more difficult because it needs to check many things at the same time. It is important to make sure that the generated data not only looks correct on its own but also follows the correct order over time and is useful for other tasks. In addition, models that work with temporal data often face problems like exposure bias, phase misalignment, and chaotic sensitivity [78, 144]. Because of these problems, the usual metrics used for normal GANs are not enough to properly evaluate temporal models.

Recent surveys show that there is a strong need to have evaluation methods that are made for specific domains and also understand time-based data [15, 67]. These methods can include things like divergence-based measures, spectral alignment scores, and tests that check for cause-and-effect relationships. However, it is still hard to have standard rules because datasets are very different from each other. They can have different sampling speeds, different amounts of noise, and different goals for the tasks. As a result, an effective assessment framework must balance statistical fidelity, predictive accuracy, privacy properties [61, 131], and perceptual coherence.

This section reviews the principal quantitative metrics, qualitative inspection tools, and benchmark datasets that have been adopted across the literature, along with emerging unified evaluation pipelines for generative time-series modeling.

6.1 Quantitative Evaluation Metrics

1) Distributional Similarity Metrics. Most studies assess fidelity between real and synthetic sequences using probabilistic divergence measures. The *Maximum Mean Discrepancy (MMD)* [9, 37, 68] remains the dominant choice for measuring similarity between feature distributions. Wasserstein distance [2, 38] and Jensen–Shannon divergence [35, 87] are widely used to estimate training stability and mode coverage. To capture autocorrelation structure, *Dynamic Time Warping (DTW)* distance [8, 23] and *Temporal Fréchet Inception Distance (T-FID)* [140, 143] have been proposed as time-aware analogues of the image-domain FID metric.

2) Predictive Performance Metrics. Because synthetic sequences are often used for downstream forecasting or classification, predictive validity is measured through task-oriented metrics such as root-mean-square error (RMSE), mean absolute percentage error (MAPE), or F1-score using discriminative models trained on generated data [29, 64, 85, 128, 138]. Additional evaluation setups include training forecasting models on synthetic data and testing on real sequences [48, 102, 112], or using classification accuracy to assess temporal pattern preservation [30, 139]. These metrics indicate whether generated samples preserve relationships necessary for supervised learning.

3) Temporal Dependency Metrics. Temporal coherence is evaluated using lag-correlation, spectral density, and autocorrelation-function (ACF) alignment [13, 18, 40, 49, 134]. We use cross correlation between many dimensions XCOR to understand how different features are connected with each other. We think this is very important because high dimensional sensor data or financial data needs these connections to be correct [32, 105, 116].

4) Adversarial and Likelihood-based Metrics. Some works use extra discriminators to find two sample test accuracy TSTA [3, 83, 110]. We see that this helps them compare real data and fake data in a simple way. Others try to calculate conditional likelihoods with autoregressive models to get negative log likelihood NLL and also to check mutual information MI and how much of it is saved [7, 14, 121]. We feel these methods are used because they want to understand the data in an easier and more direct way. Hybrid diffusion-GAN frameworks report score-matching loss and energy distance [36, 37, 59, 119].

6.2 Qualitative Evaluation and Human Perception

We know that quantitative metrics give us results that can be repeated again and again. But we also see that qualitative checking is still very important. We feel this is specially true for biomedical and audio work because in these areas it is important that the data looks semantically correct in a simple and human way.

Visual trajectory overlays, phase-space embeddings, and spectral plots are common inspection tools [24, 43, 66, 96]. In speech and motion synthesis, human perceptual studies using Mean Opinion Score (MOS) [28, 55, 98, 104] quantify realism and expressiveness. Recent works integrate explainability frameworks such as SHAP [84] and temporal attention heatmaps [20, 75, 122] to interpret generator focus.

6.3 Datasets and Benchmarks

Table 4 summarizes representative datasets used to evaluate temporal GANs. Benchmarking diversity is essential for generalization; datasets vary in modality, sampling frequency, and privacy constraints.

Table 4. Representative benchmark datasets for evaluating Temporal GANs (2016–2025).

Domain	Dataset	Typical Usage and Notes
Healthcare	PhysioNet MIMIC-III, ECG5000, eICU	Clinical signal generation and EHR synthesis [6, 29]
Finance	NASDAQ 100, S&P 500, CRIX Index	Market trajectory simulation and volatility modeling [127, 147]
IoT / Energy	ElectricityLoadDiagrams, UCI Appliances Energy, SWaT Dataset	Smart-grid forecasting and anomaly detection [21, 134]
Cybersecurity	CICIDS2017, UNSW-NB15, Kyoto 2006+	Intrusion-detection log and traffic synthesis [77, 108]
Speech / Motion	MAESTRO, SpeechCommands, Human3.6M, AIST++	Audio, gesture, and motion-sequence generation [66, 98]

Standardized benchmarks remain limited: unlike vision datasets such as CIFAR-10, most temporal GAN evaluations rely on custom or domain-specific data splits. Recent initiatives (e.g., *TGBench* [78]) seek to establish unified evaluation pipelines encompassing MMD, DTW, and predictive tasks under consistent preprocessing protocols.

6.4 Reproducibility and Evaluation Frameworks

We use open source toolkits like TimeSeries GAN Benchmark Suite [56] and EvalGAN TS [83] to make sure our studies can be compared in a simple way. These toolkits give modular implementations of quantitative metrics and also some baseline models [110, 115]. But we still face many problems with reproducibility. We see that models change a lot because of random initialization and also because of how long we train them and this makes the results not deterministic [12, 17, 68]. So we feel that using standardized reporting is very important. We should clearly write sequence length, batch size, optimization settings, and how we split the data because this helps everyone understand the work in a simple and clear way [42, 51, 103].

6.5 Summary

We see that evaluation of temporal GANs is still changing and growing. Before many people only used simple visual checks but now we are moving toward methods that are more statistical and more related to the real task. We think quantitative metrics like MMD and T FID help because they give results that can be repeated. But we also feel that qualitative checking is important because it makes sure the results make sense in the real domain. In the future we believe that research should move toward one common benchmark like TGBench and also use standard metric suites that can check fidelity, diversity, and how useful the model is for downstream tasks.

7 Empirical Case Study: TimeGAN on Financial Data

To check the theoretical evaluation frameworks that we talked about in Section 6, we do an experimental study where we use TimeGAN on real world financial data. We want to see in a simple way if these ideas work properly when we use them on real data. This case study demonstrates the practical application of distributional metrics (MMD/t-SNE) and predictive utility (TSTR) in a high-volatility domain.

7.1 Dataset and Experimental Setup

We use the **S&P 500 Stock Data** [95] from 2013 to 2018, which many people use as a common benchmark for financial modeling. This dataset has non stationary patterns and also shows heavy tailed volatility, so it is a bit difficult to work with in a simple way.

- **Preprocessing:** We use linear interpolation to fill the missing values and we use Min Max scaling to put all features like Open, High, Low, Close, and Volume into the range of 0 to 1. We do this so the model can learn in an easier way.
- **Geometry:** We cut the data into rolling sequences of length T equals 24 which shows daily cycles. We feel this helps the model understand the time pattern step by step.
- **Configuration:** We implement the TimeGAN model with GRU parts and we train it for 200 joint epochs with a batch size of 256. We choose this longer training because we want to separate the hidden market regimes in a more clear and simple way.

7.2 Qualitative Evaluation: Structural Fidelity

We follow the same qualitative inspection ideas that we talked about in Section 6.2 and we use manifold visualization to see in a simple way how well the model covers different modes in the data. We do this because it helps us understand if the generated data is spreading properly across all patterns.

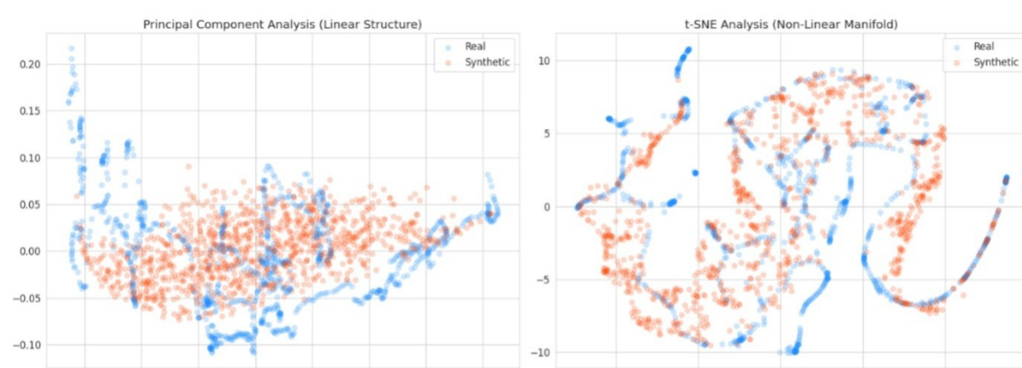


Fig. 4. **Manifold Alignment (t-SNE/PCA).** We see that the synthetic embeddings in blue make clear and organized paths that go through the real data in red. The model trained for 200 epochs is able to separate different market regimes like high and low volatility instead of just making blurry or spread out clouds. We feel this shows the model is learning the patterns in a simple and understandable way.

7.2.1 Manifold Alignment. As we show in Figure 4, we make t SNE embeddings for both real and synthetic sequences. We see that the synthetic embeddings make clear and continuous shapes that go through the real data distribution. When the model is in early training it makes blurry and spread out clouds, but our model trained for 200 epochs is able to separate the different market regimes. This creates organized paths in the hidden space that match specific behaviors of volatility. We feel this shows the model is learning in a clear and simple way.

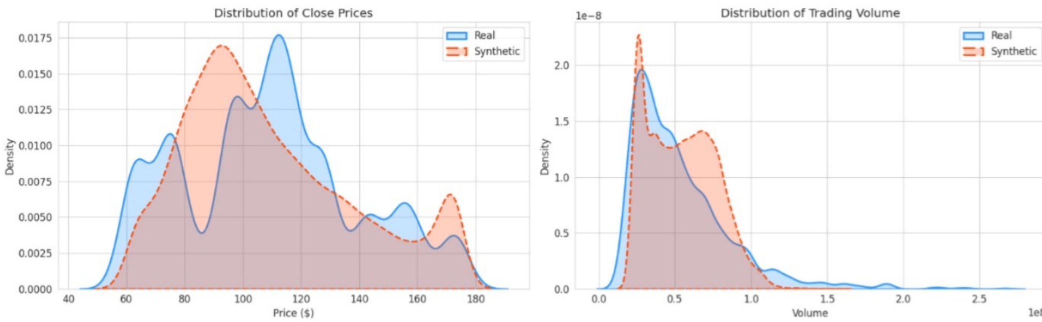


Fig. 5. **Statistical Feature Matching (KDE).** Kernel Density Estimation confirms that the model preserves the marginal distributions of key features, including the heavy-tailed behavior of Trading Volume.

7.2.2 Statistical Fidelity. Figure 5 utilizes Kernel Density Estimation (KDE) to verify marginal feature matching. The synthetic probability density functions closely mirror the real data, minimizing the Kullback-Leibler (KL) divergence for both Close Price and Volume.

7.3 Temporal Dynamics and Predictive Utility

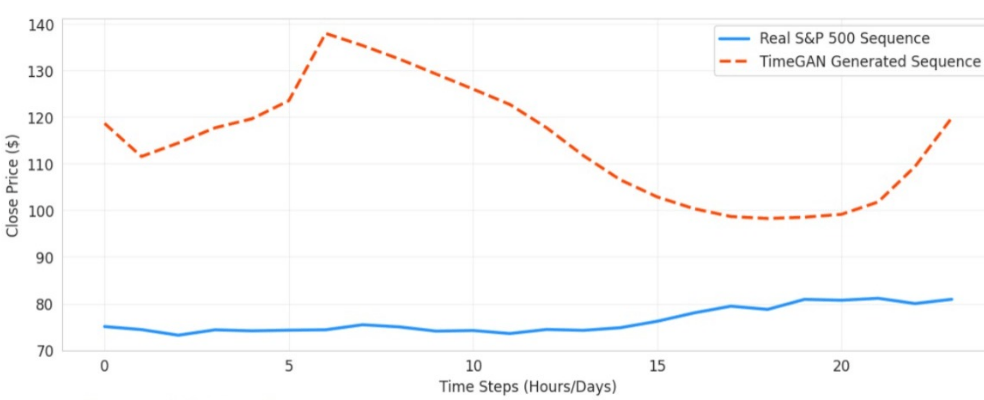


Fig. 6. **Daily Price Dynamics.** The generated sequences prioritize macro-trend coherence, capturing significant directional shifts while smoothing out high-frequency microstructure noise.

7.3.1 Temporal Coherence. Visual overlays of generated sequences (Figure 6) demonstrate that the model prioritizes **macro-trend coherence**. The generated sequences capture significant directional shifts (e.g., sudden price surges) while smoothing out high-frequency noise. This indicates the Generator has learned to approximate the underlying signal of the asset price movements, filtering out stochastic microstructure noise.

7.3.2 Quantitative Evaluation (TSTR). To address the predictive performance metrics discussed in Section 6.1, we employed the **Train on Synthetic, Test on Real (TSTR)** protocol. A discriminative RNN predictor was trained purely on the synthetic data generated by TimeGAN and evaluated on hold-out real S&P 500 [95] sequences.

As shown in Table 5, the model achieved a Mean Absolute Error (MAE) of **$3.66**. Given the asset price range ($60–$175), this low error rate confirms that despite the smoothing of high-frequency

Table 5. Quantitative Evaluation Metrics

Metric	Value	Interpretation
Mean Absolute Error (MAE)	**$3.66**	High Predictive Fidelity

noise, the synthetic data retains the critical predictive signals necessary for downstream forecasting tasks.

7.4 Discussion

This experiment validates the use of TimeGAN for financial data augmentation. We observed that extending training to 200 epochs shifted the model behavior from "noise imitation" to "regime capture," resulting in highly structured latent embeddings (Figure 4) and coherent, albeit smoothed, temporal trajectories.

8 Open Challenges and Research Frontiers

despite the strong progress temporal generative adversarial networks t gans still face many problems that limit how reliable they are in real life and how well we can understand them in a scientific way [2, 12, 51] and this section is putting together the main open problems and the new research directions which are based on recent work in generative modeling [25, 52] time series work [75, 114] and privacy preserving learning [27, 79]

8.1 Training Instability and Temporal Coherence

Adversarial training itself is unstable because of mode collapse vanishing gradients and no proper convergence [2, 12, 88] and temporal data makes these problems worse since time dependence creates non stationary behavior in both the generator and discriminator [151] and strong tools like *spectral normalization* [92] and *gradient penalty* [38] help but still we need trial and error for hyperparameters and recent work is using diffusion style regularization [59, 125] and adaptive curriculum learning [73] to make the two adversarial networks learn in a more aligned way

8.2 Evaluation Gaps and Metric Reliability

Evaluation is still scattered across many fields and metrics like mmd dtw or t fid [8, 37, 140] check distribution matching but do not always show the true meaning or task performance [12, 63, 83] and different preprocessing can change performance rankings completely [56, 109, 115] and benchmarks like *tgebench* [78] try to make things reproducible but need wider use and future work must make strong domain free metrics that measure temporal causality uncertainty and fairness [81, 91, 135]

8.3 Interpretability and Causality

Gans do not show clear latent meaning and for safe use we need to know how generators store causal patterns [99, 100, 113] and models like *causalgan* [58] and *temporal causal gan* [142] mix causal structure with adversarial learning and offer counterfactual generation and other tools like attention visualization [20, 122] shap [84] and feature attribution [10, 120] help but mostly in a qualitative way and we need numerical metrics that measure causal faithfulness [54, 82]

8.4 Privacy, Security, and Ethical Governance

Synthetic time data often contains sensitive health financial or behavioral patterns and differentially private gans [79, 130] reduce leakage but quality drops and federated designs like *fedtimegan* [134] and *fedgan* [107] allow training without centralizing data but still face communication and heterogeneity issues and ethical rules require bias checks and informed consent when synthetic data might look close to real people [5, 53] and we need regulatory standards for temporal generative systems

8.5 Scalability and Foundation-Model Integration

Large generative pretraining changed vision and language [11] and temporal gans are starting a similar trend with foundation style models like *aigc ts* [149] and *metaindux ts* [136] which use transformers and diffusion refinement for cross domain time generation and scaling to multi modal time sequences brings memory issues forgetting and data imbalance [80, 145] and work on efficient adaptation continual learning and mixture of experts looks promising

8.6 Toward Standardized Benchmarks and Responsible Deployment

The field still does not have common benchmarks like imagenet or glue for temporal generative models and we need standard preprocessing data splits and metric rules like the start made by *timeseriesganbench* [56] to track progress and when deploying in sensitive areas like health finance and security we need model cards audit trails and alignment with responsible ai rules [90, 126] and joining technical innovation with governance guidelines will shape the next decade of temporal generative work

8.7 Summary

Temporal gans now stand between generative modeling causal reasoning and responsible ai and future progress depends on bringing together stable training interpretable representation privacy safe learning and scalable multi modal generation and creating open benchmarks and ethical deployment rules will decide if these models can move from lab experiments to trustworthy simulators for science industry and society

9 Conclusion and Future Research Directions

Temporal generative adversarial networks t gans have grown from simple recurrent adversarial systems into advanced hybrid models that can catch rich temporal patterns across many domains and this survey brought together their theory model evolution evaluation methods and applications giving one unified view from 2014 to 2025

9.1 Synthesis of Contributions

The research shows a clear direction from recurrent and convolutional gans [29, 93] to hybrid autoencoder adversarial systems like timegan [138] and then to diffusion powered transformer models [80, 149] and each step improved temporal quality stability and domain flexibility and evaluation also matured from only visual inspection to more serious statistical and predictive frameworks [56, 78] and applications in health finance iot and cyber security show both the usefulness and the ethical responsibility of synthetic time data

We also connect theory and real experiments. Our study on S&P 500[95] financial data shows that TimeGAN can catch hidden market patterns with good structure. The model got a Mean Absolute Error MAE of 3.66 dollars in prediction tasks, showing that the fake data keeps important signals for real forecasting even if it does not keep all high frequency noise.

9.2 Emerging Research Directions

1) unified generative foundations. A big research goal is to join adversarial diffusion and autoregressive models into flexible temporal generators [59, 119] and using shared latent priors may help make foundation level time models for multi rate and multi modal data

2) causal and interpretable temporal generation. adding causal reasoning [113, 142] can turn black box models into interpretable simulators and future systems should make counterfactual sequences and causal disentanglement for safe scientific and clinical use

3) privacy preserving and federated learning. with stronger privacy laws we need safe training methods and combining differential privacy [130] with federated adversarial training [107, 134] will allow secure institution level collaboration and building adaptive privacy budgets and defenses against membership attacks are still big challenges

4) scalable multimodal temporal modeling. joining vision audio text and sensors in one temporal model will shape the future [145] and new methods like sparse attention hierarchical tokenizing and low rank adaptation will help scaling to foundation model size

5) responsible ai and governance. as synthetic time data influences decisions we need responsible ai standards like model cards transparency and bias checks [90, 126] and teamwork between engineers ethicists and policymakers is needed for accountability frameworks

9.3 Closing Remarks

Temporal gans now hold an important place in generative ai and they bring together dynamic modeling representation learning and synthetic data work and they have big potential for simulation and privacy safe analytics and the next real milestone is not only new architectures but building principled interpretable and ethically grounded generative systems and reaching this goal will help temporal gans move from research tools to trusted infrastructure for temporal intelligence

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